Distinguished Lecture Series I: Shing-Tung Yau, "What is a Geometric Structure"

Terence Tao · 15 May 2007

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The final Distinguished Lecture Series for this academic year at UCLA was started on Tuesday by Shing-Tung Yau. (We've had a remarkably high-quality array of visitors this year; for instance, in addition to those already mentioned in this blog, mathematicians such as Peter Lax and Michael Freedman have come here and given lectures earlier this year.) Yau's chosen topic is "Geometric Structures on Manifolds", and the first talk was an introduction and overview of his later two, titled "What is a Geometric Structure." Once again, I found this a great opportunity to learn about a field adjacent to my own areas of expertise, in this case geometric analysis (which is adjacent to nonlinear PDE).

As usual, all inaccuracies in these notes are due to myself and not to Yau, and I welcome corrections or comments. Yau's slides for the talk are available here.

Yau's first talk discussed the modern developments (mostly in the last 30 years) in geometric analysis; this is a massive subject, and to give the flavour of the field Yau presented just a few sample results from geometric analysis in this talk, mostly relating to establishing various types of geometric structures in (smooth) topological manifolds, and in Riemann surfaces in particular. (On a more personal level, Yau recalled that some of his early work in this subject, such as the proof of the Calabi conjecture, was carried out while here was a postdoc right here at UCLA.)

Geometric analysis is, by definition, the application of methods from analysis (in particular, from nonlinear PDE) to study both local and global geometry, although it turns out that in addition to analysis, methods from algebraic geometry and representation theory are also very powerful and important. Modern geometry is now a vast subject, for instance extending beyond its traditional roots in real and complex surfaces to arithmetic surfaces (as discussed earlier on this blog), and also interacting very fruitfully (and bi-directionally) with modern physics such as general relativity and string theory.

The famous and influential Erlangen program of Klein (which was further refined by Cartan) proposed to define, understand, and study geometry via the group of symmetries which preserved the structures in that geometry. In the case of the classical geometries on symmetric spaces (Euclidean geometry on $\mathbb{R}^n$, affine geometry on $\mathbb{R}^n$, projective geometry on $\mathbb{RP}^n$, spherical geometry on S^n, hyperbolic geometry on $\mathbb{H}^n$, complex geometry on $\mathbb{C}^n$, etc.), the symmetry group was a classical Lie group (e.g. the group of rigid motions, or of projective transformations, etc.). Thus for instance the notion of an angle between two intersecting lines was a concept in Euclidean geometry (because it was invariant under rigid motions) but not in affine or projective geometry (because it was not invariant under affine or projective transformations). In less symmetric situations, such as on general manifolds, the relevant symmetry groups can become infinite dimensional (e.g. the group of biholomorphic maps, or gauge transformations), or might only applicable locally (e.g. manifolds in which the local coordinate transformations between overlapping charts are in a special class such as complex, affine, unimodular, etc.); nevertheless, the basic idea of studying geometry via the symmetry group remains fundamental to the modern approach to the subject.

With this viewpoint, the only co-ordinate changes in a geometry which one should permit are those which preserve some specific algebraic structure in that geometry (e.g. complex structure, affine structure, conformal structure, projective structure, symplectic structure, or foliated structure). For instance, on complex manifolds, one should only consider co-ordinate changes which are holomorphic.

Yau's chosen topic was the construction of "geometric structures" in a given class of topological objects (manifolds, bundles, connections, maps, etc.), with everything assumed smooth for simplicity. What "geometric structure" means is a little vague, but one representative type of geometric structure is a "special" atlas of local coordinate charts on a manifold, which (locally) reduce a general geometry to a canonical geometry (such as one of the classical geometries mentioned earlier). For example, a complex structure on an (almost complex) n-dimensional manifold allows one to represent the geometry locally by the classical complex geometry $\mathbb{C}^n$.

A basic problem is then to determine whether such geometric structures actually exist for any given manifold, or class of manifolds (e.g. a topological class, conformal class, etc.). Some necessary conditions can be extracted by observing that the special geometric structure usually induces some sort of natural connection ∇ on the relevant bundle (usually the tangent bundle, though in affine or projective geometry it is the affine bundle which is important). (For instance, if the structure is a Riemannian metric, one has the Levi-Civita connection; if one has a special coordinate system which locally trivialises the bundle and whose coordinate transformations are in a special class, then the pullbacks of these trivial connections can be glued together to form a natural connection on the original manifold; and so forth.) There is a "global" condition arising from non-trivial loops in the manifold, which (by parallel transport) induce a holonomy map from a fiber of the bundle to itself, giving rise to a homomorphism from the fundamental group to the structure group. The image of this homomorphism is the holonomy group, and it must be contained in a special subgroup in order for the structure to exist; this is a key appearance of algebra in this subject. The most well-known example is that of an orientable structure on a manifold, which can only exist (i.e. the manifold is orientable) if and only if the holonomy group of any connection lies in the group of orientation-preserving linear mappings on the fiber.

Another more "local" necessary condition (which for some of these connections can be viewed as a fancy form of Clairaut's theorem $\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} = \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_i}$ from undergraduate calculus), is that the connection ∇ needs to be torsion-free if it is to arise from a special geometric structure.

One can hope that these two necessary conditions are in fact sufficient to recover (or *integrate*) the structure from its connection (which is kind of like a derivative of the structure); in other words, a geometric structure exists on a manifold if and only if one can construct a torsion-free connection with a specified holonomy group. In the real-analytic category C^ω, this is indeed the case (i.e. all such connections are *integrable*), thanks to the Cartan-Kähler theorem, which was established using the Cauchy-Kovalevskaya theorem from nonlinear PDE as the main analytic ingredient and the theory of exterior differential forms as the main geometric ingredient. This classical result already shows the importance of nonlinear PDE in differential geometry.

In the smooth category C^∞, though, things are not so simple, even at the local level, because there are additional obstructions to solvability (even local solvability) of nonlinear PDE in this category (e.g. the Lewy example). Nevertheless, by using more advanced results from nonlinear PDE one can still get results of Cartan-Kähler type, for instance the celebrated Newlander-Nirenberg theorem achieves this for complex structures, asserting that an almost complex 2n-dimensional manifold can be given a complex structure iff there exists a torsion-free connection on the tangent bundle (with the almost complex structure, of course) with holonomy group contained inside U(n).

When one wants to study more global structures, where the holonomy map plays a non-trivial role, the situation is significantly more complicated, leading to such topics as the deformation theory of such structures (e.g. the Kodaira-Spencer deformation theory in complex geometry) on the one hand, and the rigidity theory of symmetric spaces (e.g. Mostow-Margulis rigidity) that arises when the fundamental group is an infinite discrete group on the other.

For global algebraic structures, such as algebraic varieties, two powerful methods to study these structures are Torelli's theorem (which reconstructs an algebraic variety from the periods of holomorphic forms) and geometric invariant theory or GIT (studying a moduli space via the ring of invariant functions), originally introduced by Mumford in algebraic geometry, but in recent times has been shown to be

intimately linked to symplectic geometry via the theory of moment maps and symplectic reductions. One particularly important notion from GIT is that of a *stable bundle*; roughly speaking, these are bundles which have higher "slope" than any of their sub-bundles. It appears that many rigidity phenomena can be explained from the fact that actions of non-compact groups tend to ensure the stability of various bundles. Yau proposed, based on several model examples (e.g. Donaldson's theory of Hermitian Yang-Mills connections), that the solvability of various geometric PDE was intimately tied to the stability of whatever the algebraic structure was which "defined" that PDE in some sense. For instance Yau conjectured that a Fano manifold admitted a Kähler-Einstein metric if and only if it was stable in the GIT sense. The construction of solutions to geometric PDE is important as it creates canonical ways to view various algebraic and topological structures, which has proven to be a powerful tool to solve problems in algebraic geometry or topology.

Yau then focused the remainder of his talk on a quick tour of the various geometric structures on Riemann surfaces (complex manifolds of one complex dimension, or two real dimensions; not to be confused with Riemannian surfaces (Σ, g), which are two-dimensional manifolds with a Riemannian metric g). Historically, this was the first setting in which this paradigm of understanding geometry through canonical geometric structures was carried out. Indeed, Riemann's celebrated uniformisation theorem shows that given any two-dimensional compact Riemannian surface, there exists a conformally equivalent metric which has constant Gauss curvature K (this is the only curvature invariant in two dimensions; the Ricci and Riemann tensors both essentially collapse to K); if the surface is oriented this turns the Riemannian surface into a Riemann surface. One can thus view this constant curvature metric as the canonical geometric structure for this conformal class. The moduli space of all such canonical structures with a fixed topology can then be described relatively easily (it is very similar, though not quite identical to, the Teichmüller space for that manifold; the latter also requires one to mark some points on the manifold.) A version of this theory also works for finitely connected Riemannian surfaces with boundary, in which the canonical structure now acquires a boundary made of finitely many circular arcs. In principle, these moduli spaces completely describe the conformal geometries for that topology. [Yau commented that the analogous theory in higher dimension for Kähler geometries (with Ricci flatness or Einstein metrics presumably taking the place of constant curvature) remains poorly understood.]

When the surface has genus greater than one, the canonical geometric structure here has constant *negative* curvature (i.e. the geometry is hyperbolic), and Poincaré showed in fact that there is exactly one structure in this conformal class of constant curvature -1 (say). One can use this canonical structure to then induce a Riemannian metric structure (the Weil-Petersson metric) on the moduli space of conformal classes (viewed now as the space of hyperbolic metrics), because infinitesimal deformations of such metrics can be regarded as quadratic holomorphic forms $f(z)dz^2$, whose magnitude can then be computed in an L^2 sense to obtain the Riemannian structure. This metric (together with some other geometric objects, in particular an explicit functional which is geodesically convex with respect to this metric) can then be used to obtain more structure on the moduli space, for instance that its universal cover is contractible and is a Stein manifold.

Quadratic holomorphic forms also show up in other areas of geometry. For instance, (energy-minimising) harmonic maps from one Riemann surface to another generate a quadratic holomorphic form on the domain by pulling back the metric on the target. The L^2 size of these forms then can be used to construct a distance function between surfaces, which was used by Wolf to compactify Teichmüller space (which had also been achieved earlier by Thurston using a different method).

Yau also mentioned a nice appearance of quadratic holomorphic forms in Einstein's equations of general relativity in vacuum in 1+2 dimensions; one can show that initial data (very roughly speaking, the initial "position" and "velocity" of spacetime) for these equations (which must obey the constraint equations) can be identified (after a conformal change of metric) with a pair $(\Sigma, f(z)dz^2)$, where Σ is a (conformal class of a) Riemann surface and $f(z)dz^2$ is a quadratic holomorphic form; in short, initial data can be identified with a (co-)tangent vector in Teichmüller space. Then Einstein's equations

are just geodesic flow in the Weil-Petersson metric! [Sadly, there does not seem to be a similarly nice interpretation of the Einstein equation in 1+3 dimensions. In 1+2 dimensions, the Ricci tensor controls the whole curvature tensor, and so Einstein spacetimes are completely flat; of course, this is not the case in 1+3 dimensions.]

The Weil-Petersson metric is not complete (geodesics exit the boundary of Teichmüller space in finite time). There is another metric, the Teichmüller metric, on this space, which is complete, and is constructed using extremal quasiconformal maps. Somewhat weirdly, this metric is related to the negative of the Ricci tensor of the Weil-Petersson metric, as the latter generates a metric equivalent to the former (a result of Liu, Sun, and Yau). This metric is also equivalent to the canonical Kähler-Einstein metric for this moduli space (which Yau will speak on in later lectures).

The above discussion was focused on the intrinsic properties of Teichmüller space (or of moduli space of Riemann surfaces). Yau then turned to the extrinsic properties, and more precisely the question of whether one can represent this space explicitly inside complex Euclidean space. Unfortunately, Harris and Mumford showed that for sufficiently high genus (24 and higher!), the moduli space is of general type, and thus has no "good" birational representation (I was not sure exactly what this meant). On the other hand, there is a standard embedding of Teichmüller space of genus g into $\mathbb{C}^{3g-3}$, namely the Bers embedding (also discovered independently by Ahlfors). Unfortunately, this embedding is not explicit, and not well understood. But one can show that any embedding must have non-smooth boundary by the following argument of Cheng and Yau. Suppose for contradiction that one had a smooth embedding into Euclidean space. One can easily find a boundary point where the boundary is locally convex (e.g. by taking the point which is furthest away from the origin). An asymptotic analysis of the canonical Kähler-Einstein metric near this point then implies that the metric must be asymptotically constant negative curvature here; but Mumford showed that the moduli space of Riemann surfaces had a compactification where the divisor at infinity could not be blown down to a point. This can be shown to contradict the asymptotic constant negative curvature. (I admit I didn't precisely understand the connection here.)

It is still an open problem to obtain some concrete surfaces (for instance in $\mathbb{R}^3$) which would represent a given conformal class of Riemann surfaces; Torelli's theorem abstractly shows that a surface is determined by the periods of its holomorphic differential forms, but does not show how to explicitly construct a concrete surface from these periods in an algebraic manner (though there is an explicit transcendental construction of Andreotti).

Yau then turned from complex structures to projective structures on surfaces of higher genus; these can be viewed as atlases whose transformations between charts are projective transformations, and thus lift to a subset of $\mathbb{RP}^2$ as a universal cover. A particularly well-understood class of such structures are the *convex* projective structures, in which this subset is a convex subset of $\mathbb{RP}^2$ (or what is almost the same, as a convex subset of $\mathbb{R}^2$). Yau dwelled on the classification of these structures, as he viewed this as a good model example of the general paradigm mentioned earlier. In this case, there is a satisfactory classification that identifies such structures with a pair $(\Sigma, g(z)dz^3)$ of a Riemann surface with a cubic holomorphic differential. The recipe for going from the former to the latter relies heavily on nonlinear PDE; first one solves a Monge-Ampere type equation on the convex set, whose solution then gives rise (by taking the Hessian) to a projectively invariant metric on that set. Some non-trivial analysis is then needed to show that this metric is complete (basically, one needs gradient estimates near the boundary). One then takes a Legendre transform to convert the projective surface to an affine surface (or more precisely, an "affine sphere"). Affine geometry then provides a natural cubic form, the Pick cubic form, which becomes a cubic holomorphic form $g(z)dz^3$ when the affine surface is converted appropriately to a Riemann surface Σ. This procedure turns out to be reversible and completes the classification of convex projective structures. (Non-convex projective structures remain rather poorly understood.)

Yau then discussed some other relations between various geometric structures on surfaces, for instance describing how an "affine Kähler metric" (which are somewhat analogous to Calabi-Yau Kähler metrics) on a surface (with some points removed, in order to have non-trivial affine structure) can be

used to complexify the surface to a two-dimensional Kähler manifold; if the monodromy maps preserve some lattice structure, this manifold can be viewed as a torus bundle over the original surface, and are conjectured to serve as a canonical class of manifolds (which enjoy some special Lagrangian structure) to which Calabi-Yau manifolds can be deformed to; he speculated that this would be useful for analysing mirror symmetry. Yau also noted that there was an interesting "nonlinear Riemann-Hilbert problem" for these structures, namely how one can find an affine Kähler metric with prescribed monodromy. Yau also briefly discussed line fields on surfaces but without going into much detail, other than to comment on the significance of these fields in such diverse areas as the Thurston compactification of Teichmüller space, Hilbert's sixteenth problem, and fingerprint recognition.

Yau closed by stating that while some of the results and theory here extended to higher dimensions, the theory was much more complicated and less well understood for a number of reasons, one of which was that there were simply many more possible geometric structures that one could study.

[*Update*, May 15: thanks to the anonymous commenter for clarifying the relevance of generic type to the question of embedding a variety into complex Euclidean space.]

[*Update*, May 16: Corrected Yau's conjecture on Fano manifolds.]

[*Update*, May 17: Reworded the discussion on connections induced by special geometric structures. Also emphasised the desire for algebraic representations of surfaces.]

Comments (13)



Anonymous · 15 May 2007

If the moduli space is of general type, then in particular it is not rational, i.e. not birational to C^n. Thus it is not possible to write down a generic curve of genus g.
On the other hand, by the recent groundbreaking paper arXiv:math/0610203 now we know that varieties of general type do admit a canonical model, i.e. they are birational to a projective variety with negative first chern class.



Anonymous · 15 May 2007

> Some necessary conditions can be extracted from looking at the flat connection ∇ on the relevant bundle (usually the tangent bundle, though in affine or projective geometry it is the affine bundle which is important) induced from the special coordinate charts

What connection are you talking about? Levi-Civita? If so, at what point did you admit a metric? And surely it is not flat in general... In the case when a connection is flat, the holonomy just detects the fundamental group, but in general, it is sensitive to curvature also...

> Another more "local" necessary condition, which arises ultimately from Clairaut's theorem $\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} = \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_i}$ from undergraduate calculus, is that the connection ∇ needs to be torsion-free if it is to arise from a special coordinate system.

What is the definition of the "affine bundle"? In particular, I believed that in the absense of any other geometric structure enabling identification of bundles, that torsion can only be defined for a connection on the tangent bundle. Presumably you are claiming that a projective structure imposes such an identification?



sz · 15 May 2007

Dear Prof. Tao,
Your comment

“Torelli’s theorem abstractly shows that a surface is determined by the periods of its holomorphic differential forms, but does not show how to explicitly construct a concrete surface from these periods.”
is not right, such explicit construction do exist, it’s given by Aldo Andreotti in his 1958 paper “On a theorem of Torelli”. Roughly the periods determines an abelian variety, the Jacobians J of the Riemann surface, and integration along a based path gives the Abel embedding X->J, whose (g-1) iteration defines the theta divisor \Theta. The Gauss map (since the Jacobian is flat) defined by this theta divisor has an image which is the dual of the canonical curve in P^{g-1}. From this we can reconstruct the original Riemann surface.
Such explicit constructive Torelli however is completely missing for higher dimensions, notably for K3 surfaces for example, whose Torelli theorem is proved very indirectly.

 **Terence Tao** · 16 May 2007

Dear Anonymous #2,
From what I understand of Yau’s talk, it is the geometric structure which is generating the torsion-free connection on some canonical bundle. This structure might be a Riemannian metric (in which case the connection is the Levi-Civita connection on the tangent bundle, which is torsion-free by definition), or it might be a preferred coordinate system in which the coordinate transformations between charts are linear, affine or projective (in which case the tangent or affine bundle locally trivialises to a flat bundle on Euclidean space, and one uses the standard connection for that bundle, which is clearly torsion-free by Clairaut’s theorem and is also independent of the chart by the assumption on the coordinate transformations). The affine Kahler manifolds that Yau discussed in his talk are of this latter type.
The other example Yau mentioned was that of complex structures, i.e. local complex coordinate charts of an almost complex manifold. Here the analogue of the torsion tensor would be the Nijenhuis tensor. I presume that this tensor can indeed be interpreted as the torsion of some natural connection on some natural bundle (perhaps the complexified tangent bundle?), which is manifestly torsion-free when local complex coordinates exist, but I don’t have the details.
By affine bundle, I think Yau means the direct sum of the tangent bundle and the trivial R bundle, thus sections of the affine bundle are formal sums of a vector field and a scalar field.
Dear sz: I’ll ask Yau tomorrow about Andreotti’s construction. I gather that Yau was for some reason particularly interested in constructing representatives of a given conformal class of Riemann surfaces as embedded surfaces in R^3 rather than P^{g-1}, though I don’t exactly know why.

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[…] Distinguished Lecture Series I: Shing-Tung Yau, “What is a Geometric Structure” The final Distinguished Lecture Series for this academic year at UCLA was started on Tuesday by Shing-Tung Yau. […] […]

 **Anonymous** · 16 May 2007

“For instance Yau conjectured that a Fano manifold admitted a Kähler metric if and only if it was stable in the GIT sense.”
Fano manifolds are projective and therefore they admid a Kahler metric.
I believe that the conjecture is wether they admit a Kahler-Einstein metric.

 **Terence Tao** · 17 May 2007

Dear sz: I asked Yau to clarify what he meant by explicit reconstruction of a Riemann surface from the periods of its holomorphic forms. He said that he was interested in a concrete *algebraic* description of a surface (e.g. cut out by a set of algebraic equations); the Andreotti construction is more analytic in nature, giving a description which involves transcendental functions. In his talk, he

gave the analogy of modeling a (topological) torus $\mathbb{R}^2/\mathbb{Z}^2$ explicitly and algebraically in the familiar manner as the surface of revolution of a circle around an axis in $\mathbb{R}^3$; he said he would like to achieve something similar for, say, the conformal class of a geometric torus $\mathbb{R}^2/\Lambda$ but did not know of any "good" embedding into Euclidean space for this purpose (I don't know what Yau's exact criteria for "good" are, though). Yau was also interested in obtaining representations that could actually be worked out numerically by computers in a practical amount of time; apparently the Andreotti-type constructions are not feasible in this respect.



Distinguished Lecture Series II: Shing-Tung Yau, "The Basic Tools to Construct Geometric Structures" « What's new · 17 May 2007

[...] Yau's first lecture, Yau informally defined a "geometric structure" as an object (such as a metric or a [...]



sz · 18 May 2007

Dear Prof. Tao,

Thank you. You are so kind. I know Yau must have some specific motivations in his mind. I guess "cut out by a set of algebraic equations" must mean "polynomial calculation time" for him. Now he's talking about embedding an elliptic curve in to R^3, which of course would make Wiles horrified: "Where is the Modell-Weil group, man?"

But seriously, for a Riemann surface to explicitly find a set of algebraic equations to cut it out, is a difficult problem. For example let's assume the Riemann surface is the canonical curve in P^{g-1}, which is defined by an canonical ideal. This ideal then by Hilbert's syzygy theorem has a resolution, this means we can have a finite set of generators (i.e. equations for the curve), and a finite set of relations among the generators, and relations on relations, and so on. There is a well-known conjecture by your colleague Mark Green, about 25 years ago, which asserts this resolution is determined by whether the Riemann surface pocesses certain special divisors. But even this conjecture does not tell how to find these explicit set of equations.

Ars Mathematica » Blog Archive » Tao on Yau · 21 May 2007

[...] What is a geometric structure? [...]

Not Even Wrong » Blog Archive » Even More Stuff Than Usual · 31 May 2007

[...] there's a lot worth reading in Terry Tao reporting on a series of lectures by Yau at UCLA here, here and here. At the blog of fellow Fields Medalist Alain Connes, there's mention of on a [...]

Existence of Kähler-Einstein Metrics | Not Even Wrong · 7 February 2013

[...] expository articles about the problem by Donaldson and Tian, as well as a series of blog posts (here, here, and here) by Terry Tao based on lectures by [...]



Rex · 15 October 2013

The link to Yau's slides seems to be outdated. *[Corrected, thanks – T.]*